

# JIN IGARASHI

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Senior software engineer specializing in GIS, information technology, water, and sanitation fields with over 13 years of work experience in Japan and Eastern Africa. Holds an MSc (Eng) in Water, Sanitation, and Health Engineering degree with distinction from the University of Leeds, United Kingdom.

During my master's dissertation, I researched the full lifecycle costs of urban sanitation systems under the CACTUS (Climate and Costs in Urban Sanitation) project funded by the Bill and Melinda Gates Foundation. A case study for Narok town, Kenya was conducted to estimate the cost of the urban sanitation system over the sanitation value chain. Furthermore, completed an MSc in Computer Science Artificial Intelligence at the University of York from January 2024 to August 2026 as a part-time student.

Began my career developing various GIS applications at Fujitsu in Japan. Subsequently, worked in Eastern Africa (Kenya and Rwanda) on water asset management and Non-Revenue Water (NRW) management as a GIS specialist in Japan International Cooperation Agency (JICA) projects. During this time, I established a sustainable approach to maintaining geospatial applications for managing water assets by developing open-source software on GitHub. I maintain personal partnerships with water utilities in Kenya and Rwanda to continue supporting their WASH efforts as a volunteer.

Transitioned back to the software engineering industry after JICA projects, focusing on developing geospatial software in various companies in Japan before being enrolled at the University of Leeds for my master in 2021. I worked for UNDP from December 2021 to August 2025. In UNDP, I developed new web-based open-source geospatial applications to assist non-geospatial skilled UNDP staff and policymakers in easily visualizing and analyzing spatial datasets. Since September 2025, I started new role at Fracta to develop and maintain software for water utilities mainly in USA and Japan.

I am passionate about driving innovation in the WASH field due to my background in information technologies and WASH experience.

## EXPERIENCE

SEPTEMBER 2025 – PRESENT

**SOFTWARE ENGINEER**, FRACTA, JAPAN

Fracta is a software company using artificial intelligence to develop a service to detect the risk of water and sewer pipeline maintainability in the world wide (mainly in USA and Japan).

I am working in the engineering team of Fracta, responsible to develop new features, maintain the system in full stack as a software engineer.

**Skills:** Mapbox, NodeJS (JavaScript), Python, VueJS, Django, PostgreSQL/PostGIS, AWS

**DECEMBER 2021 – AUGUST 2025**

**FULL STACK GIS DEVELOPER**, UNITED NATIONS DEVELOPMENT PROGRAMME,  
NEW YORK

Develop GeoHub platform (a part of Data Future Exchange project) in SDG integration team, Bureau for Policy and Programme Support, UNDP and work remotely from the United Kingdom (until October, 2024) and Japan (from November, 2024).

I worked as a part-time GIS consultant from December 2021 to September 2022 (Individual consultant), and started working in full time from September 2022 to March 2023 (IPSA 9 Short term). Currently, I renewed my contract as a full-stack GIS developer (IPSA 9) since August 2023. In UNDP, I am mainly involved in developing GeoHub application, which is new open-source software to provide advanced data visualization, analysis and sharing tools to non-geospatial skilled UNDP staff and policymakers.

**Skills:** Maplibre, Mapbox, NodeJS (JavaScript), Python, svelte/sveltekit, PostgreSQL/PostGIS, titiler, pg\_tileserv, STAC, Azure (AppService, Blob storage, PubSub, Kubernetes, etc).

**AUGUST 2021 – MARCH 2022**

**SOFTWARE ENGINEER**, GEOLONIA INC., JAPAN

I part-time work remotely with Geolonia to develop Open Source GIS applications. I was mainly involved in open source software of open reverse geocoding and unit/charites (maplibre styling)

**Skills:** Maplibre, geocoding, reverse geocoding, Map styling

**JUNE 2021 – NOVEMBER 2021**

**SOFTWARE ENGINEER**, OCEAN EYES CO., LTD, JAPAN

Enhanced their existing WebGIS system and developed new features as a freelancer. My first work was to transfer it from Leaflet to Mapbox on Vuejs, then developed various new features on the website. Also, I established CI/CD to manage source code repository easily and sustainably.

**Skills:** NodeJS (JavaScript), Mapbox, Leaflet, VueJS

**OCTOBER 2020 – AUGUST 2021**

**SOFTWARE ENGINEER**, MIERUNE INC., JAPAN

Mainly develop WebGIS systems by using FOSS4G (Free & Open Source Software for Geospatial) as GIS specialist. Also, joined several open source GIS activities. Experience Photon (open source geocoder), Elasticsearch, Django, etc.)

**Skills:** AWS, Google Cloud, Kubernetes, Geocoding, ElasticSearch, Django, Python, etc.

**JANUARY 2020 – SEPTEMBER 2020**

**SOFTWARE ENGINEER**, ONE COMPATH CO., LTD., JAPAN

Developed and operated the one of most popular web map service in Japan which is called “MAPION”. I developed and improved their operations of mapping system which is using Mapbox Vectortiles and Mapserver on Amazon Web Service (AWS).

**Skills:** AWS, Mapbox, PostgreSQL/PostGIS

#### **APRIL 2017 – DECEMBER 2019**

##### **CONSULTANT**, KOKUSAI KOGYO CO., LTD., JAPAN

Conducted the followings projects funded by Japan International Cooperation Agency (JICA).

- The Project for Strengthening Operation and Maintenance of Rural Water Supply Systems in Rwanda (RWASOM project) (06/2018-12/2019)
- Survey for the Establishment of the Experimental Field for the GNSS System Development in the Kingdom of Thailand (11/2018-11/2019)
- Preparatory Survey for the Project for Strengthening Nzove-Ntora Principal Water Transmission Pipeline in Kigali City, Rwanda” (10/2017-11/2018)

Conducted training workshops for data collection and sharing with FOSS4G applications, then designed & created GIS database and data sharing platform for rural water supply management in RWASOM project. Collaborated with Rwandan colleagues to complete mapping all of water supply systems in the country. The system in Rwanda uses PostGIS, Lizmap/QGIS Server, QGIS Desktop, QField and Garmin GPS.

**Skills:** geospatial data collection, EPANET, water supply, design/plan water infrastructure, etc.

#### **APRIL 2014 – SEPTEMBER 2016**

##### **GIS SPECIALIST**, NAROK WATER AND SEWERAGE SERVICES CO., LTD., KENYA (JICA)

Worked as a GIS specialist funded by JICA and conducted the following activities.

- Mapped the entire water distribution network (approximately 300km length of pipeline in Total) in Narok and Ololulung’a town (Trimble GPS, ArcPAD, QGIS).
- Designed and developed GIS database & WebGIS (PostGIS, Mapserver, Leafletjs). The system links to billing system, it helps improve their revenue collections.
- Created the DMAs (District Metered Areas) for analyzing and improving Non-Revenue water management.
- Conducted training to transfer technical knowledge to Kenyan colleagues, then GIS system is continuously being used up to now since it has been developed.
- Consulted GIS work for 7 water services providers (Nyeri water, Kabarnet water, Kapsabet water, Nakuru water, Nakuru rural water, Naivasha water and Mavoko water) in Kenya.

**Skills:** Data collection for water infrastructure, WebGIS, PostgreSQL/PostGIS, EPANET, etc.

#### **APRIL 2009 – MARCH 2017**

##### **SOFTWARE ENGINEER**, FUJITSU SOCIAL SCIENCE LABORATORY, LTD., JAPAN

Designed and developed the following systems. Most of the systems are related to GIS. I left the job while I was in Kenya between April 2014 and September 2016, then back to the company.

- the development of HD map for autonomous vehicle (ArcGIS Server, Python, Django REST framework).
- the system of mapping and simulating location of ships’ movement in Singapore (Leafletjs).
- Web-GIS system for the Real Property Registration system in Ministry of Justice of Japan (Mapserver, OpenLayers, Oracle).

- An Android application for health care management (Java).
- An area marketing system by using MapInfo.
- Logistics and distribution management system with Dijkstra routing algorithm (VB.NET, SQL Server, MS Access).

**Skills:** VB.NET, JAVA, OpenLayers, Leaflet, MapServer, ArcGIS Server, SQL Server, MS Access, Oracle, etc.

## EDUCATION

**AUGUST 2026**

**MASTER OF SCIENCE**, UNIVERSITY OF YORK, YORK, UNITED KINGDOM

Certified MSc in Computer Science with Artificial Intelligence. Studied Experiment of building footprints detection in Sentinel 2 using models trained with high-resolution aerial photos in Japan as an MSc thesis.

**NOVEMBER 2022**

**MASTER OF SCIENCE (ENG)**, UNIVERSITY OF LEEDS, LEEDS, UNITED KINGDOM

Certified MSc (Eng) Water, Sanitation and Health engineering passed with distinction. Studied the full lifecycle costs of urban sanitation systems in Narok town in Kenya as an MSc dissertation.

**MARCH 2009**

**BACHELOR OF ARTS**, TOKAI UNIVERSITY, KANAGAWA, JAPAN

Studied history of international relationship between China, Korea and Japan during 16 centuries.

**JULY 2007**

**CHINESE LANGUAGE**, SCHOOL OF CHINESE AS A SECOND LANGUAGE, PEKING UNIVERSITY, BEIJING, CHINA

Studied Chinese language and Chinese history for a year.

## RECENT TALKS AND PUBLICATIONS

**SEPTEMBER 2026**

Presented the following presentation and workshop at FOSS4G 2026 at Hiroshima, Japan ([2026.foss4g.org](https://2026.foss4g.org)).

- Terra Draw – bring drawing feature to all map applications
- MAPME Initiative: Open Geospatial Resources for Development Impact
- Terra Draw Workshop – bring drawing feature to your map applications

**NOVEMBER 2025**

Presented the following presentation and workshop at FOSS4G 2025 at Auckland, New Zealand ([2025.foss4g.org](https://2025.foss4g.org)).

- maplibre-gl-terraDraw - new drawing plugin for maplibre-gl-js
- Terra Draw – cross-platform drawing library for all map applications workshop

## **JULY 2025**

Presented the following presentation at FOSS4G Europe 2025 at Mostar, Bosnia-Herzegovina ([2025.europe.foss4g.org](https://2025.europe.foss4g.org)).

- maplibre-gl-terradraw - new drawing plugin for maplibre-gl-js

## **JULY 2024**

Presented the following presentation at FOSS4G Europe 2024 at Estonia ([2024.europe.foss4g.org](https://2024.europe.foss4g.org)).

- Leave no one behind - UNDP GeoHub. Spatial data visualization and analytics for policy making

## **NOVEMBER 2023**

Igarashi, J., Evans, B., Sleight, A., Tarkash, D. N., Kennedy, R., Rosenberg, R., & Zakaria, F. (2023). Does “Low Cost” Urban Sanitation Exist? Lessons from a Global Data Set. *Environmental Science & Technology*. <https://doi.org/10.1021/acs.est.3c05731>

## **JULY 2023**

Presented the following presentations at FOSS4G 2023 Prizren, Kosovo ([2023.foss4g.org](https://2023.foss4g.org)).

- UNDP's one stop shop for cloud based geospatial data visualisation and analytical tool
- Development of maplibre applications in sveltekit

## **OCTOBER 2022**

Published Shit-flow diagram (SFD) Lite report for Narok town, Kenya at [sfd.susana.org](https://sfd.susana.org). The report shows the latest sanitation situation in the municipality.

## **AUGUST 2022**

Co-presented “Rural water supply mapping by using FOSS4G application in Rwanda” at FOSS4G 2022 Firenze, Italy.

## **FEBRUARY 2022**

Presented about “A case study of Open Source GIS in rural water management in Rwanda” at Webinar for water utilities of West African countries hosted by JICA.

## **OCTOBER 2021**

Presented three talks about use cases of Rwanda and Kenya’s GIS activities in FOSS4G 2021 Buenos Aires Online (<https://2021.foss4g.org>).

- “FOSS4G software developments for Water Utilities Management in Eastern Africa by using Vector Tiles”
- “The impact of application of FOSS4G on Non-Revenue Water” (Presenter was Mr. James Njenga, I was backup)
- “An implementation of FOSS4G - QGIS, QField and Vector Tiles for rural water supply management in Rwanda” (Presenter was Mr. Moise Irankunda, I was backup)

## **FEBRUARY 2021**

Joined “Open Education for QGIS” session by Hans van der Kwast (lecturer of IHE Delft Institute) as a panel at QGIS Open Day.

#### **JANUARY 2021**

Presented about “A case study of Open Source GIS in rural water management in Rwanda” at Webinar for water utilities of West African countries hosted by JICA.

#### **NOVEMBER 2020**

Presented about “Use cases of Vector Tiles implementation for water utilities in Eastern Africa” at FOSS4G Korea 2020 online conference.

#### **OCTOBER 2020**

Presented about “A cheap and easy way to utilize GIS for effective management of water utility” at WaQuAc Net (Water Quality Asian Cooperation Network, <http://waquac.net/english> ) Webinar.

#### **SEPTEMBER 2020**

Presented about “Vector Tiles implementation for water services providers in Eastern Africa” in FOSS4G Hokkaido.

#### **AUGUST 2019**

Presented about use cases of Rwanda and Kenya’s GIS activities in FOSS4G 2019 Bucharest (<https://2019.foss4g.org>).

- URL of Kenya’s presentation: <https://media.ccc.de/v/bucharest-135-case-study-of-non-revenue-water-management-of-narok-water-in-kenya>
- URL of Rwanda’s presentation: <https://media.ccc.de/v/bucharest-30-case-study-of-data-collection-data-sharing-for-rural-water-supply-management-in-rwanda>

## **QUALIFICATIONS**

#### **NOVEMBER 2022**

**MSc (Eng) Water, Sanitation and Health Engineering** (School of Civil Engineering, University of Leeds). Passed with distinction.

#### **APRIL 2022**

**Designing a Multidimensional Poverty Index** (United Nations Development Programme (UNDP) and the Oxford Poverty and Human Development Initiative (OPHI))

#### **NOVEMBER 2021**

**BSAFE** (United Nations Department of Safety and Security)

#### **NOVEMBER 2021**

**Prevention of sexual exploitation and abuse (PSEA)** (UNICEF)

#### **NOVEMBER 2021**

**Prevention of sexual harassment and abuse of authority (2021 version)** (UN & UN agencies)

#### **JUNE 2017**

**PCM MONITORING EVALUATION** (FASID - FOUNDATION FOR ADVANCED STUDIES ON INTERNATIONAL DEVELOPMENT)

**MAY 2017**

**PCM PARTICIPATORY PLANNING** (FASID – FOUNDATION FOR ADVANCED STUDIES ON INTERNATIONAL DEVELOPMENT)

**NOV 2009**

**FUNDAMENTAL INFORMATION TECHNOLOGY ENGINEER EXAMINATION** (INFORMATION TECHNOLOGY PROMOTION AGENCY, JAPAN)

**MAY 2009**

**INFORMATION TECHNOLOGY PASSPORT EXAMINATION** (INFORMATION TECHNOLOGY PROMOTION AGENCY, JAPAN)

## **LANGUAGES**

- English (Fluent, IELTS Overall 6.5, CEFR: B2, in March 2021)
- Japanese (Native)
- Chinese (Fluent, HSK Chinese Proficiency Test Grade 9 in April 2009)
- Swahili (Fluent)